

matter where assigned, and SL can be thought of as defiers, who will transfer no matter where assigned. The values of the principal strata are *not* affected by assignment of home hospital type—which value $[T(\ell)$ or $T(s)]$ is observed *is* affected by treatment assignment, but the bivariate values are not, and therefore $(T(\ell), T(s))$ is, formerly, a partially observed covariate.

Now, we consider what is called the “monotonicity” assumption or the “no-defier” assumption—that is, we assume that the SL principal stratum is empty. In our setting, this assumption is very plausible, and because it excludes the SL principal stratum, we have only three principal strata: LL , LS and SS . Under this assumption, the possible principal strata for each *observed* combination of home hospital type and treating hospital type in each propensity subclass are shown in the fourth columns of Tables 3–7. The observed $\ell \rightarrow S$ group (the second row in Tables 3–7) must be composed of SS patients because they can be neither LL nor SL patients, respectively—because they were assigned ℓ but treated in S and therefore are not LL patients, and there are no SL patients by the monotonicity assumption. Similarly, the observed $s \rightarrow L$ group (the third row of Tables 3–7) must be LL patients because they were assigned s but were treated in L .

In contrast, the observed $\ell \rightarrow L$ subgroup (the first row of Tables 3–7) could be compliers, and so be in LS , or noncompliers who are members of the LL principal stratum (who were assigned to home hospital type L , and to which they would have transferred for their treating hospital type if they were assigned to a small home hospital type). Hence, we split row 1 into two sub-rows in the fourth column of Tables 3–7. Similarly, the observed $s \rightarrow S$ subgroups (the fourth row of Tables 3–7) could be compliers, and so be in LS , or noncompliers who are members of the SS principal stratum, and so is also split into two sub-rows.

We can approximate the proportion of patients in each principal stratum, as shown in the fifth columns of Tables 3–7. More explicitly, from the second row of Table 7, columns (1) and (3), we see that 2/20 are observed to be $\ell \rightarrow S$. Because of the assumed random assignment into ℓ and s within propensity score subclasses, we have that approximately 10% of the patients belong to the principal stratum SS , as shown in the fifth column of Table 7. Similarly, from the third row of Table 7, columns (1) and (3), we infer that approximately $6/9 \approx 67\%$ of patients belong to principal stratum LL in this subclass, as shown in the fifth columns of Table 7.

Hence, we can approximate the fraction of compliers, the LS principal stratum in this subclass, by simple subtraction: $100\% - 10\% - 67\% = 23\%$. The sixth column in Table 7 indicates the approximate number of LS patients in each of the four rows of observed patients. Analogous calculations are summarized in Tables 3–6 for the other propensity score subclasses. Even if we could perfectly identify all the LS patients, which we cannot, the sample sizes are small, and so inference for the causal effect of treating

hospital when it equals home hospital will be imprecise. Nevertheless, we outline the planned analysis in Section 5.6 because these are the only data available to study this question. Importantly, we anticipate that in each subclass there are some compliers who are treated in large volume hospitals and some compliers who are treated in small volume hospitals.

5.5. *ITT and CACE = ITT_{LS} and their estimation.* The average causal effect of home hospital type on survival is the comparison of the potential survival outcomes of all N patients under $h_i = \ell$ and under $h_i = s$,

$$\text{ITT} = \frac{1}{N} \sum_{i=1}^N [Y_i(\ell) - Y_i(s)],$$

where ITT is the Intention-To-Treat (ITT) effect of the assignment of large versus small home hospital type. Under unconfounded assignment of home hospital type, we are able to estimate ITT by taking the average observed difference in Y for large volume hospital patients and small volume hospital patients within each propensity subclass, weighting each subclass-specific estimate by the total number in that subclass and averaging the estimates. Because we are not examining survival outcome data at this design stage, we cannot calculate these estimates, but we saw this approach in the breast cancer example of Section 4. In this problem using the template of a randomized block experiment with noncompliance, the estimation is more subtle.

When $G_i = LS$, the home hospital type equals the treating hospital type, that is, $h_i = T_i$. The causal effect of home hospital type in the LS principal stratum is defined to be

$$\text{CACE} \equiv \text{ITT}_{LS} = \frac{1}{N_{LS}} \sum_{i \in LS} (Y_i(L) - Y_i(S)),$$

where N_{LS} is the number of LS patients, and CACE means “Compliance Average Causal Effect” [Imbens and Rubin (1997)]. ITT_{LS} can be interpreted as either the intention-to-treat effect of home hospital type for complying patients or the intention-to-treat effect of treating hospital type for complying patients, because for the LS principal stratum, $h_i = T_i$. Under monotonicity, the LS principal stratum is the only stratum of patients where we can learn about the causal effects of treating hospital type because the patients in the other principal strata, LL and SS , will always be exposed to the same treating hospital type.

CACE is easily estimated once we identify the individuals in the LS stratum, and we have not yet identified any particular member of the $\ell \rightarrow L$ or $s \rightarrow S$ rows (rows one and four) in Tables 3–7 as being in the LS principal stratum, and so we cannot yet compare average outcomes in this stratum. Nevertheless, we can find a unique method-of-moments estimate

of the causal effects of assigned (= treating) hospital type within the LS principal stratum under, what are considered, medically very justifiable assumptions, which in general are called “exclusion restrictions.” The resulting estimator of CACE is known as the “instrumental variable estimate” [Angrist, Imbens and Rubin (1996)]. Better (e.g., Bayesian) methods of estimation exist [e.g., see Imbens and Rubin (1997)].

5.6. *Exclusion restrictions.* There are two exclusions restrictions. The first exclusion restriction is for patients in the LL principal stratum. It states that, for all $i \in LL$, $Y_i(\ell) = Y_i(s)$, that is, there is no effect on potential outcomes Y of being assigned to a large (ℓ) versus small (s) home hospital type for patient $i \in LL$. The medical justification for this restriction is that patient i would be treated in a large hospital type (L) under either assignment, and one’s medical outcome is considered a result of where one is treated not where one is diagnosed. The exclusion restriction for patients in the SS principal stratum is analogous; for all $i \in SS$, $Y_i(\ell) = Y_i(s)$, that is, for those patients who would be treated in a small hospital type (S) whether assigned to l or s , there is no effect of assignment on the Y potential outcomes.

Now, ITT for all patients can be written as

$$\text{ITT} = \pi_{LS}\text{ITT}_{LS} + \pi_{SS}\text{ITT}_{SS} + \pi_{LL}\text{ITT}_{LL},$$

where π_{LS}, π_{SS} and π_{LL} are the fractions of the sample, and ITT_{LS} , ITT_{SS} and ITT_{LL} are the intention-to-treat effects in the LS , SS and LL strata, respectively. Because the exclusion restrictions force ITT_{SS} and ITT_{LL} to be identically zero, this equation becomes

$$\text{ITT} = \pi_{LS}\text{ITT}_{LS},$$

or

$$\text{ITT}_{LS} = \text{ITT}/\pi_{LS}.$$

Thus, the instrumental variables estimate of the ITT effect of treating hospital among compliers is found by dividing the estimated ITT effect of home hospital type by the estimated fraction of the sample in LS .

The planned analysis will use Bayesian versions of this estimator within each propensity score subclass, and then average over all subclasses. An initial Bayesian analysis that partially benefits from the propensity score analysis presented here, but also involves data on stomach cancer patients, is presented in Rubin et al. (2008).

6. Discussion. This article advocates the position that observational studies for causal effects need to be designed to approximate randomized experiments. This enterprise requires careful thought and execution, and not simply running mindless regression programs and looking at coefficients. In most situations, this design effort will be more intellectually demanding than a similar effort for an analogous randomized experiment. Of critical importance, final outcome data cannot be used in design without compromising the objectivity of the study design. Propensity score methods are extremely helpful tools for reconstructing the underlying hypothetical experiment that lead to the observed data. Sometimes, the hypothetical approximating randomized experiment is one with complications, such as noncompliance, and then the principal stratification framework can be extremely helpful. But most important is for the worker in observational studies to stay focused on approximating a plausible hypothetical underlying randomized experiment.

A final comment concerns the application of this perspective to actual randomized experiments, especially those with covariates that have not been used in the randomization (e.g., not used to create blocks). In such cases, we would expect random imbalances in some covariates, and if there is concern that these covariates may be related to outcomes, the application of the techniques described here, with no access to final outcome data, preserves the objectivity of the experiment, whereas model-based adjustments, unless fully specified a priori, would compromise that objectivity. This approach has been applied, for example, in a study of school vouchers [Barnard et al. (2003)] and in a study of vertical disease transmission during delivery [Zell et al. (2007)].

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